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Team Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

Description :

This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch** field is set to **EEE**, while administrators retain full access.

Timer :

3 hrs

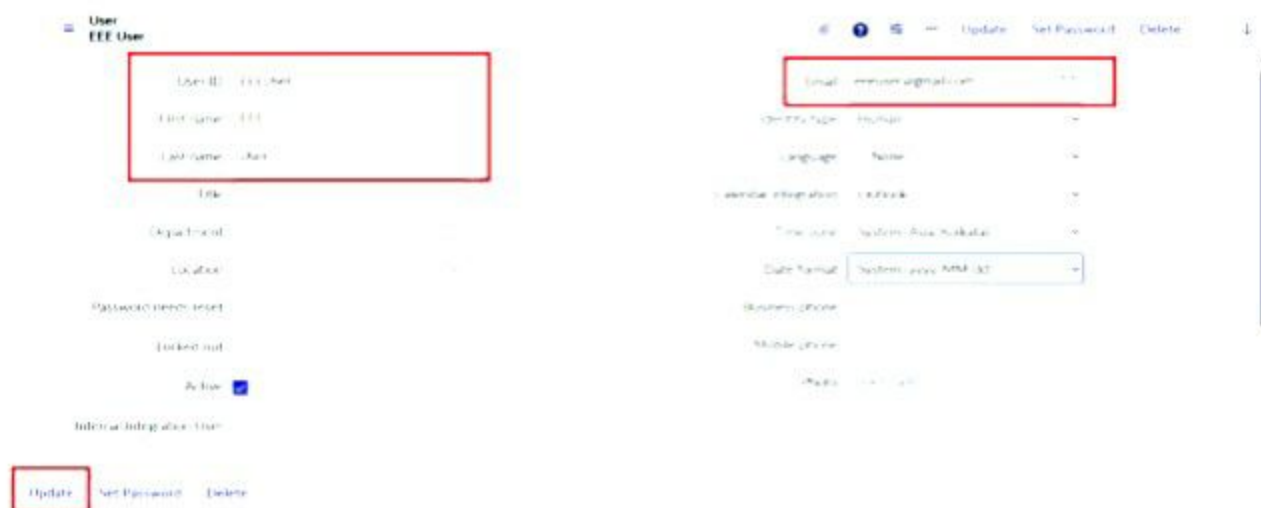
Milestone-1 Creation of Users and Roles

User Creation:

1. Log in to the ServiceNow instance with **admin** access.
2. **Create** a test user:
 - Navigate to User Administration → Users
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record



GO FURTHER



Role Creation

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as **bb1**
- **Description**: Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the **EEE User**



Role 001

Name 001

Application Global

Elevated privileges ☐

Description

Update Delete

[SNLMSI Versions \(1\)](#)

Tables Creation

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false



3. Click **Submit**.

4. Add the following fields to the **u_institution_details** table:

- **Student Roll Number**– Auto Number
- **Student Name** – Reference- User
- **Faculty Name** – Reference- User
- **Branch** – Choice (ECE, EEE, CSE)
- **Email**– String
- **Phone Number**– String
- **Description**– Multi String

5. Save the Table

Table Columns for text Search 1 to 13 of 13

Dictionary Entries

Column label	Type	Reference	Max length	Default value
Updates	Integer	(empty)	40	
Created by	String	(empty)	40	
Student Roll Number	String	(empty)	40	javascript:global.getTextObj(Number Padded):
Description	String	(empty)	1,000	
Updated by	String	(empty)	40	
Phone Number	String	(empty)	40	
Sys ID	Sys ID (GUID)	(empty)	32	
Created	Date/Time	(empty)	40	
Faculty name	Reference	User	32	
Branch	String	(empty)	40	
Updated	Date/Time	(empty)	40	
Student Name	Reference	User	32	
Email	String	(empty)	40	

Insert a new row...

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

Institution Details Student Roll Number Search Actions on selected rows...

Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
Search	Search	Search	Search	Search	Search
INS0001006	Praneeth	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan Kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	Praneeth	Srinu Nandipati	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tutor	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tutor	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Creation of Access Control Lists(Read ACL)

1. In the **Application Navigator**, search for **System Security** → **Access Control (ACL)**.
2. Elevate "Security_admin" role
3. Click **New** to create a new ACL.
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true

- **Advanced:** true

5. In the **Requires role** related list, add

- bbl (custom role)

6. Add Data condition as **Branch is EEE**.

7. In the **Script** field, add the following script.

```
(function () {
```

```
// Allow admin users full access
```

```
if (gs.hasRole('admin')) {
```

```
    return true;
```

```
}
```

```
// Allow only EEE branch users to see EEE records
```

```
if (gs.hasRole('bbl')){
```

```
    return true;
```

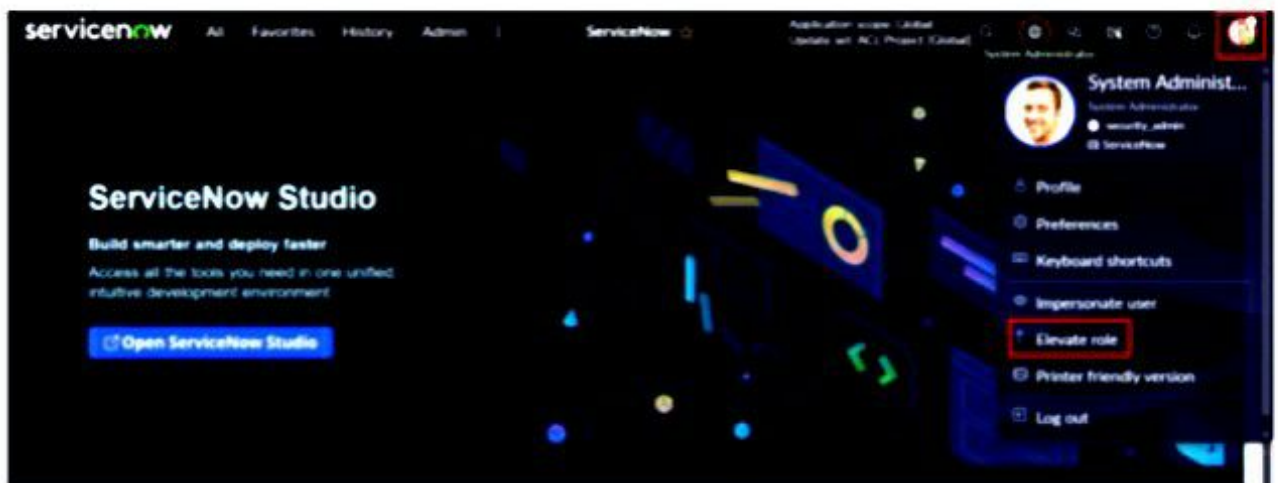
```
}
```

```
// Deny access for all others
```

```
return false;
```

```
}()};
```

8. Click **Submit** to save the ACL.



Elevate role

Close dialog

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel



Access Controls Updated							
Search							
Actions on selected rows...							
New							
All							
☐	Name	Decision Type	Operation	Type	Active	Updated by	Updated
☐	Search	Search	Search	Search	Search	Search	Search
☐	sn.try.build.agent.message	Deny Unless	read	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.freemium.usage.log	Deny Unless	read	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.event.telemetry	Deny Unless	report.view	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.conversation	Deny Unless	report.view	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.task.telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.task.telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.event.telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.task.telemetry	Allow If	read	record	true	system	about 1 month ago
☐	sn.try.build.agent.knowledge.source	Allow If	write	record	true	system	19-10-2025 22:23:04
☐	sn.try.build.agent.event.telemetry	Allow If	report.view	record	true	system	19-10-2025 22:23:04

Access Control

u_institution_details

Update

Delete

Type

record

Operation

read

Decision Type

Allow If

Admin overrides

☐

Application

Global

Active

☒

Advanced

☒

Protection policy

None

Name

Institution Details (u_institution_details)

None

Description

Applies to

No. of records matching the conditions: 6

Add Filter Condition

Add OR Clause

choose field

oper

value

Data Condition

Condition

No. of records matching the conditions: 2

Add Filter Condition

Add OR Clause

Branch

is

EEE

AND

OR

Advanced Condition

Controlled by References

☐

Script

Run on ECMAScript 2021 (ES12) mode

1

2

3

4

5

6

7

8

```

(function() {
  if (gs.hasRole('001')) {
    return true;
  } else {
    return false;
  }
})();

```

9. Impersonate **EEE** users and open the Student Records list.
10. Copy the table name `u_institution_details.list/LIST` and paste in the All menu.
11. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1** role can view only **EEE** branch records

- User without the role cannot view any records

Institution Details

Student Roll Number

Search

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

Search

Search

Search

Search

Search

Search

Search



No records to display

- Admin user can view **all** records regardless of branch

Institution Details

Student Roll Number

Search

Actions on selected rows...

New

All

Student Roll Number

Faculty name

Student Name

Branch

Email

Phone Number

Search

Search

Search

Search

Search

Search

INS0001006

ITIL User

Praneeth

ECE

praneeth@gmail.com

8989090906

INS0001008

Mella Sravan kumar

Abraham Lincoln

CSE

sravan@gmail.com

8978908989

INS0001005

ITIL User

Srinu Nandipati

ECE

srinu@gmail.com

9089909090

INS0001007

Abel Tuter

Guna Sai

EEE

saiguna@gmail.com

9090898765

INS0001001

Abraham Lincoln

Yaswanth Nikku

CSE

yaswanth@gmail.com

8989898989

INS0001004

Abel Tuter

Prashanth Reddy

EEE

prashanth@gmail.com

9090909090

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL.(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record



Access Controls Updated							Search	+ - 🔍 🔗 📄 📁	New
Name	Decision Type	Operation	Type	Active	Updated by	Updated			
Access to build agent knowledge source	Deny Unless	read	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge source by	Deny Unless	read	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge source	Deny Unless	delete	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge source	Deny Unless	edit	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge	Deny Unless	delete	read	True	system	19-10-2025 22:23:04			
Access to build agent named telemetry	Deny Unless	export, allow	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge	Deny Unless	export, allow	read	True	system	19-10-2025 22:23:04			
Access to build agent knowledge by	Deny Unless	export	read	True	system	19-10-2025 22:23:04			
Access to build agent build telemetry	Deny Unless	read	read	True	system	19-10-2025 22:23:04			
Access to build agent build telemetry by	Deny Unless	delete	read	True	system	19-10-2025 22:23:04			
Access to build agent build telemetry	Allow If	read	read	True	system	about 3 months ago			
Access to build agent knowledge source	Allow If	write	read	True	system	19-10-2025 22:23:04			
Access to build agent named telemetry	Allow If	export, allow	read	True	system	19-10-2025 22:23:04			

Access Control
u_institution_details

Type: record

Operation: create

Decision Type: Allow if

Admin overrides: ☒

Protection policy: -- None --

Name: Institution Details [u_institution_details] -- None --

Description: /

Applies To: [has all records matching the condition](#)

[Add Filter Condition](#) [Add OR Clause](#)

-- choose field -- -- oper -- -- value --

Application: Global

Active: ☒

Advanced: ☐

Update Delete

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
bb2
Insert a new row...

1 to 1 of 1

Security Attribute Condition

Verify :

- User with bb1 & bb2 roles can view only EEE branch records b New Button

Institution Details Student Roll Number Search

Actions on selected rows... New

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		satguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

Creation of Access Control Lists(Write ACL)

12. Click New to create a new ACL(Write)

13. Enter the following details:

- Type: record
- Operation: Write
- Name: u_institution_details

- Active: true

14. In the Requires role related list, add:

- bb3 (custom role)

15. No need to add any Data condition

16. Save the ACL Record



Access Controls Updated: Search							Actions on selected rows		New
Name	Decision Type	Operation	Type	Active	Updated by	Updated			
Search	Search	Search	Search	Search	Search	Search			
sn-fcm-build-agent-fcm-message	Deny Unless	read	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-fcm-message-log	Deny Unless	read	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-conversation-log	Deny Unless	delete	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-message-identity	Deny Unless	reject, allow	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-notification	Deny Unless	reject, allow	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-task-identity	Deny Unless	delete	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-task-identity-log	Deny Unless	read	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-task-identity-log	Deny Unless	delete	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-task-identity-log	Allow If	read	record	true	system	about 3 months ago			
sn-fcm-build-agent-knowledge-source	Allow If	write	record	true	system	19-10-2025 22:23:04			
sn-fcm-build-agent-knowledge-source-log	Allow If	reject, allow	record	true	system	19-10-2025 22:23:04			

Access Control **u_institution_details** Update Delete

Type: record

Operation: **write**

Decision type: Allow If

Admin overrides: ☒

Protection policy: -- None --

Name: **Institution Details (u_institution_details)**

Description:
 Applies to: [No. of records matching the conditions 4](#)
 Add Filter Condition Add OR Clause
 -- choose field -- -- op -- -- value --

Requires role

Role
bb3

[More info](#)

[Insert a new role...](#)

Verify :

- User with **bb1 & bb2 & bb3** roles can view and edit the **EEE branch records** along with **New Button**

Institution Details Actions on selected rows... New

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
IN50001007	EEE		com	Abel Tutor	9090898765	Cuna Sal
IN50001004	EEE		il.com	Abel Tutor	9090909090	Prashanth Reddy

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
10-10101	EEE	ACL is Executed	ajayashan@gmail.com	Ajay Shan	9876543210	Ajay Shan
10-10102	EEE		ajayashan@gmail.com	Ajay Shan	9876543210	Ajay Shan

Creation of Access Control Lists(Delete ACL)

17. Click **New** to create a new ACL (Delete)
18. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
19. In the **Requires role** related list, add:
 - bb4 (custom role)
20. No need to add any Data condition
21. Save the ACL Record



Access Controls Updated							
Search							
Name	Decision Type	Operation	Type	Active	Updated by	Updated	
sn_try_build_agent_message	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_freemium_usage_log	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_message	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_conversation	Deny Unless	report_view	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Deny Unless	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Deny Unless	read	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Deny Unless	delete	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_task_telemetry	Allow If	read	record	true	system	about 3 months ago	
sn_try_build_agent_knowledge_source	Allow If	write	record	true	system	19-10-2025 22:23:04	
sn_try_build_agent_event_telemetry	Allow If	report_view	record	true	system	19-10-2025 22:23:04	

Access Control New record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

Type: record

Operation: delete

Decision Type: Allow If

Application: Global

Active: ☒

Advanced: ☐

Admin overrides: ☒

Protection policy: -- None --

Name: Institution Details [u_institution_details]

Description:

Applies To: No. of records matching the condition: 6

Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Requires role

Role
bb4
Insert a new row...

Security Attribute Condition

Verify :

- Users with bb1 & bb2 & bb3 & bb4 roles can see, create and edit along with delete the EEE branch records.

The screenshots illustrate the steps to delete a record from a table in a web application interface.

Table Data (from first screenshot):

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE	ACL is Executed	saigana@gmail.com	Ashu Kumar	9090898765	Guna Sai
INS0001004	EEE	ACL is Executed	prashanth@gmail.com	Ashu Kumar	9090909090	Prashanth Reddy

Actions on selected rows (from second screenshot):

- Delete
- Assign tag
- Print Log
- Assign Platform Security
- Strategies - IP Address Management
- Cloud Service Management
- Basic Platform
- Security
- IT Asset Management
- IT Service Management
- IT Operations Management
- More

Confirmation Dialog (from third screenshot):

Confirmation
Delete the selected record?

Buttons: Cancel, Delete

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.